pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Resampling and the As-Of Value

WEEK 1 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

One number per period, and its date

A monthly report shows one 10 Yr yield for September, not twenty-one. Someone chose which one. Then someone wrote a date next to it.

A weekly or monthly series is labelled by the date of the observation it holds.

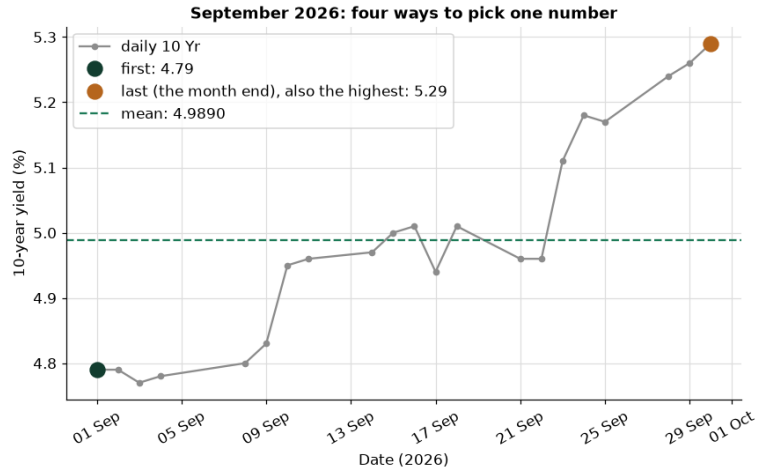
Course 1's resample("ME").last() labelled January 2015 as the 31st, a Saturday with no row. In a join, that is a trap.

No FRED key needed today: the Treasury file and the synthetic spreads.

Today

4.1	Which observation
4.2	last_per_period
4.3	Good Friday 2025
4.4	asfreq against last
4.5	value_as_of
4.6	Month ends, 2026

Which observation stands for a month



September 2026, 10 Yr (percent)

```
first: 4.79 percent, on 2026-09-01

last: 5.29 percent, on 2026-09-30

mean: 4.9890 percent

high: 5.29 percent, on 2026-09-30
```

The course's convention: the last observation of each calendar month (ME) and of each week ending Friday (W-FRI).

42 of 141 complete months end on a day with no row. In 2026: January and February (Saturday), May (Sunday).

resample labels against last_per_period

resample("ME").last()	last_per_period	10 Yr
2026-01-31	2026-01-30	4.26
2026-02-28	2026-02-27	3.97
2026-03-31	2026-03-31	4.30
2026-04-30	2026-04-30	4.40
2026-05-31	2026-05-29	4.45
2026-06-30	2026-06-30	4.44
2026-07-31	2026-07-31	4.75
2026-08-31	2026-08-31	4.75
2026-09-30	2026-09-30	5.29
2026-10-31	2026-10-02	5.28

Look up the 2 Yr on resample's labels

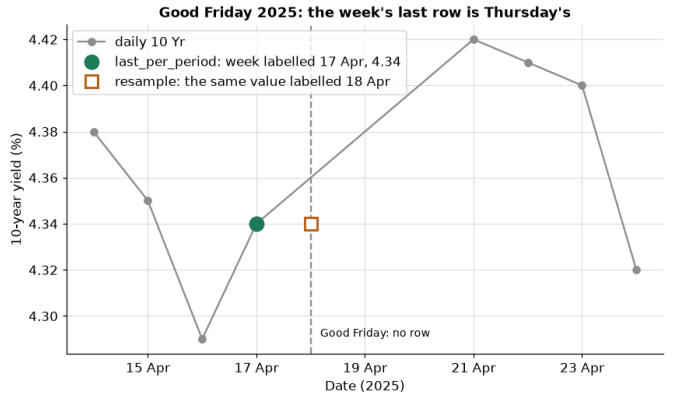
Found on 99 of 142 labels. The other 43 come back NaN, with no error.

Same value in every one of 142 months; 43 labels differ. Every last_per_period label is a market date.

October 2026: resample says 2026-10-31, after the file ends. last_per_period says 2026-10-02: a part month.

The week of Good Friday 2025

```
ten.loc["2025-04-14":"2025-04-18"]
2025-04-14    4.38
2025-04-15    4.35
2025-04-16    4.29
2025-04-17    4.34
```



20 of 614

weeks end on a Thursday: no row on that Friday

8

of those Fridays are Good Fridays; 12 are other Fridays

2026-04-03

a Good Friday the file does have a row for

The calendar comes from the file, not from a rule.

asfreq against last: a grid with holes



Two questions

last_per_period asks for the last observation in each period, and keeps a part period.

asfreq asks for the value on each grid date: NaN where there is no row, and a part period is dropped.

20 of 614

Fridays on the W-FRI grid are NaN

42 of 141

month ends on the ME grid are NaN; October 2026 is dropped

126 of 3,066

business days NaN: day 3's missing weekdays

Filling the holes puts Thursday's value on a closed Friday: display only, never before a change, a window, or a join.

The as-of value: on or before

The last observation on or before the date. Before the first observation: no answer, so stop.

```
md.value_as_of(ten, "2026-09-07")
10 Yr as of 2026-09-07: 4.78 percent, from 2026-09-04 (Friday)
as of 2026-10-03: 5.28 percent, from 2026-10-02
as of 2026-09-30: 5.29 percent, from 2026-09-30
```

pandas' asof, before the start

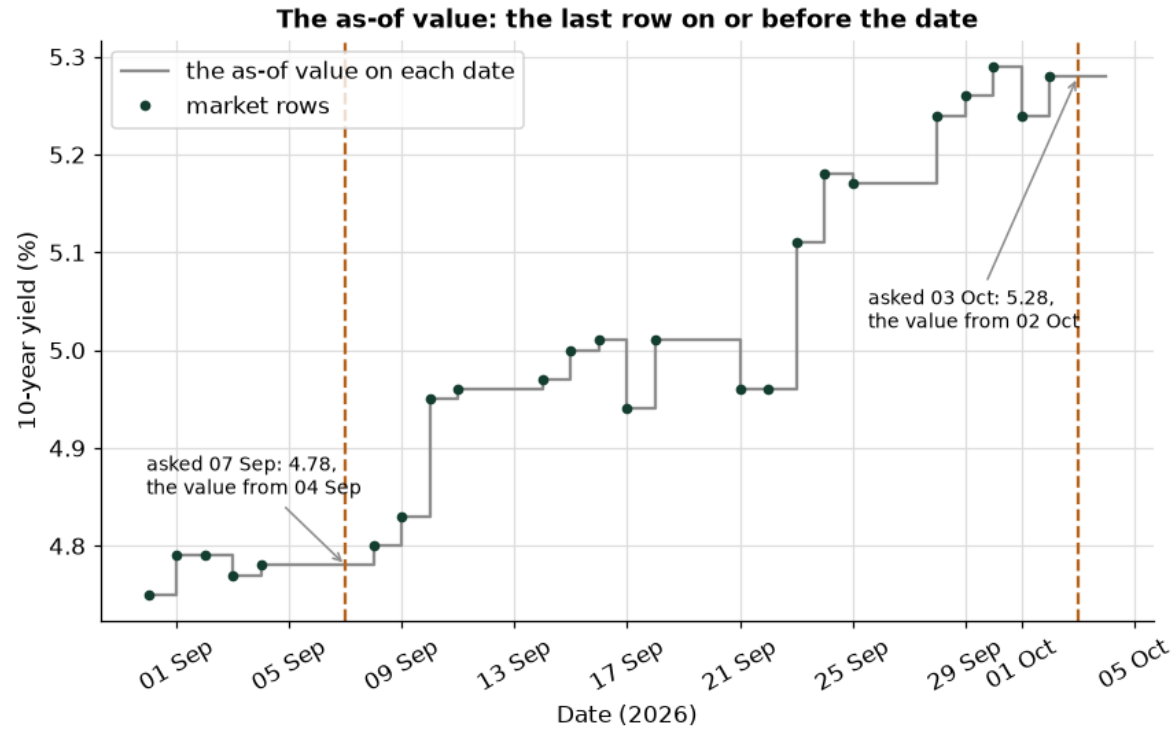
```
asof on 2014-12-31: nan
```

A silent hole, and no date returned.

```
ValueError: 2014-12-31 is before the first observation, 2015-01-02
```

Four tests: month-end label, Good Friday week, as of a holiday, before the start raises. 17 passed.

The as-of value, drawn



Asked 7 Sep: 4.78 percent, the value from 4 Sep.

Asked 3 Oct: 5.28 percent, the value from 2 Oct.

Asked 30 Sep: 5.29 percent, the value from 30 Sep.

Month-end levels for 2026

Date	10 Yr (%)	2s10s (bp)	IG spread (bp)
2026-01-30	4.26	74.0	91.3
2026-02-27	3.97	59.0	95.6
2026-03-31	4.30	51.0	96.5
2026-04-30	4.40	52.0	106.8
2026-05-29	4.45	47.0	102.2
2026-06-30	4.44	30.0	100.1
2026-07-31	4.75	47.0	97.2
2026-08-31	4.75	41.0	94.3
2026-09-30	5.29	41.0	98.8
2026-10-02	5.28	45.0	97.0

The IG spread is synthetic: invented, not ICE, Moody's, or any index.



10 Yr: low 3.97 on 27 Feb, high 5.29 on 30 Sep

2s10s (bp): low 30 on 30 Jun, high 74 on 30 Jan

IG spread (bp): low 91.3 on 30 Jan, high 106.8 on 30 Apr

Excel side by side

Excel side by side (tested in Excel)

`=EOMONTH(DATE(2026,5,15),0)`	2026-05-31, a Sunday (WEEKDAY returns 7)
`=MAXIFS(A2:A2941,A2:A2941,"<="&EOMONTH(...))`	2026-05-29
`=XLOOKUP(that date,A2:A2941,M2:M2941)`	4.45
`=A2+MOD(6-WEEKDAY(A2),7)`	2025-04-16 gives 2025-04-18
`=MAXIFS(A2:A2941,A2:A2941,"<="&DATE(2025,4,18))`	2025-04-17
`=XLOOKUP(DATE(2026,9,7),A2:A2941,M2:M2941,,-1)`	4.78 (from 2026-09-04)
`=XLOOKUP(DATE(2014,12,31),A2:A2941,M2:M2941,,-1)`	#N/A

Filled down: all 142 months and 614 weeks equal last_per_period. Pivot table by month: Max of date is the last market date; Max of 10 Yr is the month's high, not its month end in 6 of 10 months.

Tested in Excel 16.0 build 20527, 2026-11-02.

Save the day: git stash

1. Experiment

month_first added to marketdata.py, not committed

2. Stash

git stash push -m "...": the file is back to the commit

3. Work

pytest on the clean file (17 passed); hand over the table

4. Pop

git stash pop: the change is back. Read it, then decide

```
stash@{0}: On main: experiment: first observation of each month
```

first() is the wrong edge (4.79 for September 2026, not 5.29), so git restore marketdata.py discards it.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: month_end_levels promises the month's last market day and uses resample("ME").first(): 4.79 for September 2026, not 5.29.

Code that runs is not the same as code that is right.

What you can do now

- The last observation stands for a week or a month, labelled by its own date
- last_per_period: series or table, W-FRI or ME, weeks that end on a Thursday
- asfreq is a grid with holes: good for finding them, not for levels
- value_as_of: the value and its date, and a stop before the first observation
- Excel: EOMONTH, MAXIFS, XLOOKUP with match mode -1

Week 1, Day 5 Case study: caching, why licensed series are never saved, and the course's daily dataset